

Simplification with don't cares

Exercise 1

ab / cd	00	01	11	10
00	1	0	-	-
01	-	-	1	1
11	0	-	1	0
10	-	0	1	0

Exercise 2

ab / cd	00	01	11	10
00	-	1	-	0
01	-	0	-	-
11	0	1	0	-
10	1	1	-	0

Exercise 3

ab / cd	00	01	11	10
00	1	-	1	-
01	0	0	1	1
11	0	1	0	0
10	-	-	-	0

Exercise 4

ab / cd	00	01	11	10
00	1	0	0	1
01	-	-	-	1
11	0	0	1	-
10	0	-	0	-

Exercise 5

ab / cd	00	01	11	10
00	1	1	-	0
01	0	-	1	0
11	-	-	0	0
10	0	1	0	1

Exercise 6

ab / cd	00	01	11	10
00	0	1	-	1
01	-	-	1	-
11	0	1	-	1
10	0	1	-	1

Exercise 7

ab / cd	00	01	11	10
00	0	-	1	-
01	-	0	0	1
11	1	1	-	-
10	-	-	-	-

Exercise 8

ab / cd	00	01	11	10
00	-	1	0	0
01	0	-	1	0
11	0	1	0	0
10	-	0	-	1

Exercise 9

ab / cd	00	01	11	10
00	1	1	1	1
01	0	1	1	-
11	1	0	1	1
10	1	0	0	-

Exercise 10

ab / cd	00	01	11	10
00	-	-	-	0
01	-	0	0	-
11	-	0	1	0
10	-	1	0	-

Exercise 11

ab / cd	00	01	11	10
00	0	1	-	0
01	1	0	1	0
11	0	-	-	1
10	-	0	0	1

Exercise 12

ab / cd	00	01	11	10
00	0	1	-	-
01	1	-	0	1
11	-	0	1	0
10	0	0	0	1

Exercise 13

ab / cd	00	01	11	10
00	-	0	1	-
01	1	0	0	-
11	1	-	1	-
10	-	1	0	0

Exercise 14

ab / cd	00	01	11	10
00	0	1	1	1
01	-	1	0	-
11	-	-	-	0
10	0	-	1	1

Exercise 15

ab / cd	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	-	1	0	1
10	-	0	1	1

Exercise 16

ab / cd	00	01	11	10
00	1	-	0	0
01	1	1	0	-
11	0	1	0	0
10	-	0	-	-

Exercise 17

ab / cd	00	01	11	10
00	0	1	0	-
01	-	1	1	0
11	0	1	1	-
10	-	1	0	1

Exercise 18

ab / cd	00	01	11	10
00	0	1	1	-
01	1	1	1	1
11	-	0	0	1
10	1	-	-	-

Exercise 19

ab / cd	00	01	11	10
00	-	1	1	1
01	-	1	-	-
11	0	1	-	1
10	-	-	0	0

Exercise 20

ab / cd	00	01	11	10
00	-	-	-	0
01	1	0	-	-
11	-	0	-	-
10	1	0	-	1

Exercise 1

ab / cd	00	01	11	10
00	1	0	-	-
01	-	-	1	1
11	0	-	1	0
10	-	0	1	0

Solution to Exercise 1

There are 6 prime implicants:

$$\bar{a}b$$

$$\bar{a}c$$

$$bd$$

$$cd$$

$$\bar{a}\bar{d}$$

$$\bar{b}\bar{c}\bar{d}$$

A minimum cover is made of 2 prime implicants:

$$cd + \bar{a}\bar{d}$$

Exercise 2

ab / cd	00	01	11	10
00	-	1	-	0
01	-	0	-	-
11	0	1	0	-
10	1	1	-	0

Solution to Exercise 2

There are 8 prime implicants:

$$\bar{b}\bar{c}$$

$$\bar{b}d$$

$$\bar{a}bc$$

$$\bar{a}cd$$

$$a\bar{c}d$$

$$\bar{a}b\bar{d}$$

$$bc\bar{d}$$

$$\bar{a}\bar{c}\bar{d}$$

A minimum cover is made of 2 prime implicants:

$$\bar{b}\bar{c} + a\bar{c}d$$

Exercise 3

ab / cd	00	01	11	10
00	1	-	1	-
01	0	0	1	1
11	0	1	0	0
10	-	-	-	0

Solution to Exercise 3

There are 5 prime implicants:

$$\bar{a}\bar{b}$$

$$\bar{a}c$$

$$\bar{b}\bar{c}$$

$$\bar{b}d$$

$$a\bar{c}d$$

A minimum cover is made of 3 prime implicants:

$$\bar{a}c + a\bar{c}d + \bar{b}\bar{c}$$

Exercise 4

ab / cd	00	01	11	10
00	1	0	0	1
01	-	-	-	1
11	0	0	1	-
10	0	-	0	-

Solution to Exercise 4

There are 5 prime implicants:

$$\bar{a}b$$

$$bc$$

$$\bar{a}\bar{d}$$

$$c\bar{d}$$

$$a\bar{b}\bar{c}d$$

A minimum cover is made of 2 prime implicants:

$$bc + \bar{a}\bar{d}$$

Exercise 5

ab / cd	00	01	11	10
00	1	1	-	0
01	0	-	1	0
11	-	-	0	0
10	0	1	0	1

Solution to Exercise 5

There are 5 prime implicants:

$$\bar{a}d$$

$$\bar{c}d$$

$$ab\bar{c}$$

$$\bar{a}\bar{b}\bar{c}$$

$$a\bar{b}c\bar{d}$$

A minimum cover is made of 4 prime implicants:

$$\bar{a}d + \bar{c}d + \bar{a}\bar{b}\bar{c} + a\bar{b}c\bar{d}$$

Exercise 6

ab / cd	00	01	11	10
00	0	1	-	1
01	-	-	1	-
11	0	1	-	1
10	0	1	-	1

Solution to Exercise 6

There are 3 prime implicants:

$$c$$

$$d$$

$$\bar{a}b$$

A minimum cover is made of 2 prime implicants:

$$c + d$$

Exercise 7

ab / cd	00	01	11	10
00	0	-	1	-
01	-	0	0	1
11	1	1	-	-
10	-	-	-	-

Solution to Exercise 7

There are 5 prime implicants:

$$\begin{aligned}
 &a \\
 &\bar{b}c \\
 &\bar{b}d \\
 &b\bar{d} \\
 &c\bar{d}
 \end{aligned}$$

A minimum cover is made of 3 prime implicants:

$$a + \bar{b}d + c\bar{d}$$

Exercise 8

ab / cd	00	01	11	10
00	-	1	0	0
01	0	-	1	0
11	0	1	0	0
10	-	0	-	1

Solution to Exercise 8

There are 7 prime implicants:

$$\begin{aligned}
 &a\bar{b}c \\
 &\bar{a}\bar{b}\bar{c} \\
 &\bar{a}bd \\
 &\bar{a}\bar{c}d \\
 &b\bar{c}d \\
 &a\bar{b}\bar{d} \\
 &\bar{b}\bar{c}\bar{d}
 \end{aligned}$$

A minimum cover is made of 4 prime implicants:

$$\bar{a}bd + b\bar{c}d + \bar{a}\bar{c}d + a\bar{b}\bar{d}$$

Exercise 9

ab / cd	00	01	11	10
00	1	1	1	1
01	0	1	1	-
11	1	0	1	1
10	1	0	0	-

Solution to Exercise 9

There are 7 prime implicants:

$$\bar{a}\bar{b}$$

$$\bar{a}c$$

$$bc$$

$$\bar{a}d$$

$$a\bar{d}$$

$$\bar{b}\bar{d}$$

$$c\bar{d}$$

A minimum cover is made of 4 prime implicants:

$$bc + \bar{a}d + a\bar{d} + \bar{b}\bar{d}$$

Exercise 10

ab / cd	00	01	11	10
00	-	-	-	0
01	-	0	0	-
11	-	0	1	0
10	-	1	0	-

Solution to Exercise 10

There are 6 prime implicants:

$$\bar{b}\bar{c}$$

$$\bar{c}\bar{d}$$

$$\bar{a}\bar{b}d$$

$$\bar{a}b\bar{d}$$

$$a\bar{b}\bar{d}$$

$$abcd$$

A minimum cover is made of 2 prime implicants:

$$\bar{b}\bar{c} + abcd$$

Exercise 11

ab / cd	00	01	11	10
00	0	1	-	0
01	1	0	1	0
11	0	-	-	1
10	-	0	0	1

Solution to Exercise 11

There are 8 prime implicants:

$$\begin{aligned}
 &abc \\
 &abd \\
 &\bar{a}\bar{b}d \\
 &\bar{a}cd \\
 &bcd \\
 &a\bar{b}\bar{d} \\
 &ac\bar{d} \\
 &\bar{a}b\bar{c}\bar{d}
 \end{aligned}$$

A minimum cover is made of 4 prime implicants:

$$\bar{a}\bar{b}d + \bar{a}b\bar{c}\bar{d} + bcd + ac\bar{d}$$

Exercise 12

ab / cd	00	01	11	10
00	0	1	-	-
01	1	-	0	1
11	-	0	1	0
10	0	0	0	1

Solution to Exercise 12

There are 9 prime implicants:

$$\begin{aligned}
 &\bar{a}\bar{b}c \\
 &\bar{a}b\bar{c} \\
 &\bar{a}\bar{b}d \\
 &\bar{a}\bar{c}d \\
 &\bar{a}b\bar{d} \\
 &\bar{a}c\bar{d} \\
 &\bar{b}c\bar{d}
 \end{aligned}$$

$$b\bar{c}\bar{d}$$

$$a b c d$$

A minimum cover is made of 4 prime implicants:

$$\bar{b}c\bar{d} + a b c d + \bar{a}\bar{c}d + \bar{a}b\bar{d}$$

Exercise 13

ab / cd	00	01	11	10
00	-	0	1	-
01	1	0	0	-
11	1	-	1	-
10	-	1	0	0

Solution to Exercise 13

There are 6 prime implicants:

$$a b$$

$$a \bar{c}$$

$$\bar{a} \bar{d}$$

$$b \bar{d}$$

$$\bar{c} \bar{d}$$

$$\bar{a} \bar{b} c$$

A minimum cover is made of 4 prime implicants:

$$a b + a \bar{c} + \bar{a} \bar{b} c + \bar{c} \bar{d}$$

Exercise 14

ab / cd	00	01	11	10
00	0	1	1	1
01	-	1	0	-
11	-	-	-	0
10	0	-	1	1

Solution to Exercise 14

There are 7 prime implicants:

$$\bar{b} c$$

$$b \bar{c}$$

$$a d$$

$$\begin{aligned} &\bar{b}d \\ &\bar{c}d \\ &\bar{a}b\bar{d} \\ &\bar{a}c\bar{d} \end{aligned}$$

A minimum cover is made of 2 prime implicants:

$$\bar{b}c + \bar{c}d$$

Exercise 15

ab / cd	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	-	1	0	1
10	-	0	1	1

Solution to Exercise 15

There are 5 prime implicants:

$$\begin{aligned} &\bar{a}c \\ &\bar{b}c \\ &a\bar{d} \\ &c\bar{d} \\ &ab\bar{c} \end{aligned}$$

A minimum cover is made of 4 prime implicants:

$$\bar{a}c + \bar{b}c + ab\bar{c} + c\bar{d}$$

Exercise 16

ab / cd	00	01	11	10
00	1	-	0	0
01	1	1	0	-
11	0	1	0	0
10	-	0	-	-

Solution to Exercise 16

There are 6 prime implicants:

$$\begin{aligned} &\bar{a}\bar{c} \\ &a\bar{b}c \end{aligned}$$

$$b\bar{c}d$$

$$\bar{a}b\bar{d}$$

$$a\bar{b}\bar{d}$$

$$\bar{b}\bar{c}\bar{d}$$

A minimum cover is made of 2 prime implicants:

$$b\bar{c}d + \bar{a}\bar{c}$$

Exercise 17

ab / cd	00	01	11	10
00	0	1	0	-
01	-	1	1	0
11	0	1	1	-
10	-	1	0	1

Solution to Exercise 17

There are 8 prime implicants:

$$bd$$

$$\bar{c}d$$

$$abc$$

$$\bar{a}b\bar{c}$$

$$a\bar{b}\bar{c}$$

$$a\bar{b}\bar{d}$$

$$ac\bar{d}$$

$$\bar{b}c\bar{d}$$

A minimum cover is made of 3 prime implicants:

$$bd + \bar{c}d + \bar{b}c\bar{d}$$

Exercise 18

ab / cd	00	01	11	10
00	0	1	1	-
01	1	1	1	1
11	-	0	0	1
10	1	-	-	-

Solution to Exercise 18

There are 9 prime implicants:

$$\bar{a}b$$

$$a\bar{b}$$

$$\bar{a}c$$

$$\bar{b}c$$

$$\bar{a}d$$

$$\bar{b}d$$

$$a\bar{d}$$

$$b\bar{d}$$

$$c\bar{d}$$

A minimum cover is made of 3 prime implicants:

$$\bar{a}d + a\bar{d} + b\bar{d}$$

Exercise 19

ab / cd	00	01	11	10
00	-	1	1	1
01	-	1	-	-
11	0	1	-	1
10	-	-	0	0

Solution to Exercise 19

There are 5 prime implicants:

$$\bar{a}$$

$$bc$$

$$\bar{b}\bar{c}$$

$$bd$$

$$\bar{c}d$$

A minimum cover is made of 3 prime implicants:

$$\bar{a} + bc + \bar{c}d$$

Exercise 20

ab / cd	00	01	11	10
00	-	-	-	0
01	1	0	-	-
11	-	0	-	-
10	1	0	-	1

Solution to Exercise 20

There are 8 prime implicants:

$$a c$$

$$b c$$

$$c d$$

$$a \bar{d}$$

$$b \bar{d}$$

$$\bar{c} \bar{d}$$

$$\bar{a} \bar{b} \bar{c}$$

$$\bar{a} \bar{b} d$$

A minimum cover is made of 2 prime implicants:

$$a \bar{d} + \bar{c} \bar{d}$$